

Guiselle Tatiana Zambrano Penagos

Backend Software Engineer

Bogotá, Colombia

+57 312 568 1791 | gtzp.6174@gmail.com | gtzambanop | gtzambanop

Professional Summary

Full Stack Software Engineer with more than four years of experience designing and developing backend solutions for enterprise applications serving customers across Colombia and Mexico.

Experienced in Python, Flask, PostgreSQL, SQLAlchemy and REST API development, with a strong background in database design, query optimization, business process automation and scalable software architecture.

Passionate about building maintainable backend systems, solving complex technical problems and continuously improving software quality through clean design and refactoring.

Professional Experience

Autolab

Full Stack Software Engineer

October 2021 – May 2026

Bogotá, Colombia

Backend Platform

- Designed and developed REST APIs using Python, Flask and SQLAlchemy to support enterprise applications for vehicle maintenance and fleet management.
- Designed PostgreSQL schemas, optimized complex SQL queries, and developed reusable data access components, improving application performance and maintainability.
- Built reusable backend services and shared business logic that reduced code duplication and standardized development across multiple products.

Fleet Management Platform

- Designed and implemented backend services supporting fleet maintenance workflows, quotation approvals, appointment management and operational communication for enterprise customers.
- Architected modular backend components with role-based access control and scalable domain separation, enabling future service evolution and improving maintainability.

CRM Platform

- Refactored legacy backend modules by centralizing business rules, reducing technical debt and improving long-term maintainability.
- Developed backend functionality for warranties, quotations, workshop administration and operational management.
- Automated operational workflows through WhatsApp integrations, reducing manual communication between advisors, workshops and fleet customers.

Production Support

- Investigated and resolved production incidents using Sentry, SQL analysis and application logs across backend, database and frontend components.
- Performed root cause analysis and implemented long-term fixes to improve application stability and reliability.

Technical Skills

Programming

Python
JavaScript
SQL

Backend

Flask
SQLAlchemy
REST APIs
JWT/Auth

Databases

PostgreSQL
Redis
Database Design

Frontend

React
Ember.js
HTML/CSS

Tools

Docker
Git
Linux

Education

Universidad Nacional de Colombia

B.Sc. in Systems and Computer Engineering

Certifications

Universidad Nacional de Colombia

Diploma Program in Cloud-Based Software Architecture and Quality

Bogotá, Colombia

January 2026 – June 2026

Languages

Spanish Native

English B1 (Actively improving professional communication and technical fluency)